Table A3. Random parameters binary logit model with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), single-vehicle crashes without a motorcycle, Thailand, 2016–2023 (yearly periods), partially constrained temporal structure.

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Constant [serious injury / fatal]				
Constant [2016]	–1.765	–20.89		
Constant [2017–2019]	–2.049	–26.34		
Constant [2020–2021]	–2.504	–28.73		
Constant [2022–2023]	–2.752	–30.77		
Variables producing the same parameter value across all periods [2016–2023]				
Vehicle involvement				
One passenger car involved (1 if yes, 0 otherwise) [2016–2023]	0.132	4.44	–0.0112	0.0112
One 6-wheel truck involved (1 if yes, 0 otherwise) [2016–2023]	–0.343	–5.03	0.0260	–0.0260
Crash type				
Rear-end collision in the same lane (1 if yes, 0 otherwise) [2016–2023]	0.869	7.78	–0.0939	0.0939
Vehicle mounted a traffic island (1 if yes, 0 otherwise) [2016–2023]	–0.725	–12.43	0.0525	–0.0525
Roadway characteristics				
Painted median (1 if yes, 0 otherwise) [2016–2023]	–0.199	–3.20	0.0157	–0.0157
Raised or earth median (1 if yes, 0 otherwise) [2016–2023]	–0.197	–4.61	0.0160	–0.0160
Concrete pavement surface (1 if yes, 0 otherwise) [2016–2023]	–0.350	–2.41	0.0058	–0.0058
(Standard deviation of parameter distribution; 35.8% above zero)	0.962	4.11		
On grade (1 if yes, 0 otherwise) [2016–2023]	–2.699	–2.83	–0.0084	0.0084
(Standard deviation of parameter distribution; 25.8% above zero)	4.163	4.55		
T-intersection (1 if yes, 0 otherwise) [2016–2023]	–0.360	–3.87	0.0270	–0.0270
Environmental characteristics				
Wet road surface (1 if yes, 0 otherwise) [2016–2023]	–0.305	–8.22	0.0240	–0.0240
Temporal characteristics				
Songkran festival period (1 if yes, 0 otherwise) [2016–2023]	–0.478	–3.35	–0.0119	0.0119
(Standard deviation of parameter distribution; 37.8% above zero)	1.534	7.16		
Evening (1 if yes, 0 otherwise) [2016–2023]	0.105	2.90	–0.0089	0.0089
Variables with temporally varying parameters				
Vehicle involvement				
One 10-wheel truck involved (1 if yes, 0 otherwise) [2016–2022]	–0.732	–8.11	0.0511	–0.0511
One truck with more than 10 wheels involved (1 if yes, 0 otherwise) [2016–2019]	–0.919	–11.02	0.0697	–0.0697
One truck with more than 10 wheels involved (1 if yes, 0 otherwise) [2020–2023]	–0.718	–8.92	0.0436	–0.0436
Van involved (1 if yes, 0 otherwise) [2016–2019]	0.739	7.15	–0.0867	0.0867
Van involved (1 if yes, 0 otherwise) [2021–2023]	0.695	4.27	–0.0619	0.0619
Pedestrian involved (1 if yes, 0 otherwise) [2016–2022]	4.005	30.76	–0.6492	0.6492
Pedestrian involved (1 if yes, 0 otherwise) [2023]	5.108	15.40	–0.7599	0.7599
Crash type				
Ran off road to the left and hit an object (1 if yes, 0 otherwise) [2016–2018]	–0.580	–8.19	0.0512	–0.0512
Ran off road to the left and hit an object (1 if yes, 0 otherwise) [2019–2023]	–0.260	–4.52	0.0190	–0.0190
Ran off road to the right and hit an object (1 if yes, 0 otherwise) [2016–2018]	–0.535	–6.93	0.0470	–0.0470
Ran off road to the right and hit an object (1 if yes, 0 otherwise) [2019–2022]	–0.315	–4.87	0.0235	–0.0235
Ran off road to the right and hit an object (1 if yes, 0 otherwise) [2023]	–0.537	–4.96	0.0330	–0.0330
Ran off road to the left (1 if yes, 0 otherwise) [2016–2021]	0.476	7.91	–0.0482	0.0482
Ran off road to the left (1 if yes, 0 otherwise) [2022–2023]	0.774	8.09	–0.0660	0.0660
Ran off road on a right bend and hit an object (1 if yes, 0 otherwise) [2016–2018]	–0.785	–7.74	0.0638	–0.0638
Ran off road on a right bend and hit an object (1 if yes, 0 otherwise) [2021–2023]	–0.369	–3.97	0.0237	–0.0237
Ran off road on a left bend and hit an object (1 if yes, 0 otherwise) [2016–2018]	–0.743	–6.96	0.0605	–0.0605
Ran off road on a left bend and hit an object (1 if yes, 0 otherwise) [2019–2023]	–1.337	–1.91	0.0194	–0.0194
(Standard deviation of parameter distribution; 24.6% above zero)	1.949	2.67		
Ran off road to the right (1 if yes, 0 otherwise) [2016–2017]	0.669	7.33	–0.0818	0.0818
Ran off road to the right (1 if yes, 0 otherwise) [2018–2023]	0.433	5.82	–0.0384	0.0384

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Ran off road on a left bend (1 if yes, 0 otherwise) [2018]	–0.477	–2.25	0.0375	–0.0375
Ran off road on a left bend (1 if yes, 0 otherwise) [2019–2020]	0.760	4.79	–0.0798	0.0798
Ran off road on a right bend (1 if yes, 0 otherwise) [2018]	–0.394	–1.85	0.0317	–0.0317
Ran off road on a right bend (1 if yes, 0 otherwise) [2019–2023]	0.537	4.99	–0.0483	0.0483
Crash cause				
Speeding (1 if yes, 0 otherwise) [2016–2019]	0.284	4.32	–0.0259	0.0259
Speeding (1 if yes, 0 otherwise) [2020–2023]	0.681	9.05	–0.0448	0.0448
Cutting in too closely (1 if yes, 0 otherwise) [2016]	0.427	1.83	–0.0526	0.0526
Cutting in too closely (1 if yes, 0 otherwise) [2019–2023]	0.621	4.20	–0.0575	0.0575
Driver fell asleep (1 if yes, 0 otherwise) [2016–2019]	0.719	9.12	–0.0810	0.0810
Driver fell asleep (1 if yes, 0 otherwise) [2020]	1.209	9.82	–0.1327	0.1327
Driver fell asleep (1 if yes, 0 otherwise) [2021–2022]	0.663	5.90	–0.0577	0.0577
Driver fell asleep (1 if yes, 0 otherwise) [2023]	0.978	8.32	–0.0855	0.0855
Roadway characteristics				
Crash on the main carriageway (1 if yes, 0 otherwise) [2016–2020]	–0.501	–5.90	0.0405	–0.0405
Road under construction (1 if yes, 0 otherwise) [2018–2023]	0.242	2.17	–0.0204	0.0204
Depressed median (1 if yes, 0 otherwise) [2016]	–0.164	–2.06	0.0177	–0.0177
Depressed median (1 if yes, 0 otherwise) [2017–2018]	0.102	1.77	–0.0096	0.0096
Depressed median (1 if yes, 0 otherwise) [2019–2023]	–0.110	–2.45	0.0083	–0.0083
Barrier median (1 if yes, 0 otherwise) [2016–2018]	–0.583	–6.35	0.0494	–0.0494
Barrier median (1 if yes, 0 otherwise) [2019–2023]	–1.087	–16.07	0.0646	–0.0646
Horizontal curve (1 if yes, 0 otherwise) [2017–2018]	0.256	2.90	–0.0249	0.0249
Horizontal curve (1 if yes, 0 otherwise) [2019–2020]	–0.266	–3.19	0.0215	–0.0215
Sag vertical curve (1 if yes, 0 otherwise) [2016–2018]	0.322	2.22	–0.0347	0.0347
Sag vertical curve (1 if yes, 0 otherwise) [2019–2020]	–0.471	–2.00	0.0349	–0.0349
U-turn with deceleration lane only (1 if yes, 0 otherwise) [2019]	0.612	2.43	–0.0638	0.0638
Bridge section (1 if yes, 0 otherwise) [2017–2023]	0.588	5.93	–0.0566	0.0566
Environmental characteristics				
Dark, no street lighting (1 if yes, 0 otherwise) [2016–2022]	0.466	9.85	–0.0453	0.0453
Temporal characteristics				
Weekend (1 if yes, 0 otherwise) [2023]	0.201	2.72	–0.0141	0.0141
New Year festival period (1 if yes, 0 otherwise) [2018–2022]	–0.393	–2.11	–0.0253	0.0253
(Standard deviation of parameter distribution; 40.7% above zero)	1.663	5.89		
Midnight / early morning (1 if yes, 0 otherwise) [2016–2018]	0.218	4.63	–0.0220	0.0220
Midnight / early morning (1 if yes, 0 otherwise) [2019–2023]	0.135	3.41	–0.0105	0.0105
Region				
Eastern region (1 if yes, 0 otherwise) [2020–2023]	0.176	2.91	–0.0135	0.0135
Northern region (1 if yes, 0 otherwise) [2019–2023]	–0.840	–4.04	0.0108	–0.0108
(Standard deviation of parameter distribution; 30.1% above zero)	1.608	6.42		
Southern region (1 if yes, 0 otherwise) [2016–2018]	0.216	4.07	–0.0221	0.0221
Southern region (1 if yes, 0 otherwise) [2022–2023]	0.298	4.23	–0.0217	0.0217
Western region (1 if yes, 0 otherwise) [2018–2020]	0.224	1.25	–0.0590	0.0590
(Standard deviation of parameter distribution; 57.1% above zero)	1.245	3.75		
Western region (1 if yes, 0 otherwise) [2021–2023]	0.263	3.45	–0.0201	0.0201
Heterogeneity in the means of random parameters				
On grade [2016–2023]: Raining [2016–2023]	–2.456	–4.38		
Northern region [2019–2023]: Horizontal curve [2021–2023]	–0.454	–2.81		
On grade [2016–2023]: Eastern region [2016–2019]	2.108	3.04		
Heterogeneity in the variances of random parameters				
On grade [2016–2023]: Road under construction [2016–2017]	–0.672	–1.86		
Model statistics				
Number of observations			73,731	

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Number of estimated parameters		83		
Log-likelihood at zero, LL(0)		−51106.43		
Log-likelihood with constant only, LL(C)		−26810.27		
Log-likelihood at convergence, LL(β)		−23590.45		
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$		0.538		
AIC / BIC		47,346.9 / 48,111.2		
Random parameter distribution; simulation		Normal; 2,000 Halton draws		

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median, one pickup truck. Periods in brackets share one parameter value; for a variable with temporally varying parameters, periods not shown have a parameter of zero. Temporal structure from the partially constrained procedure in the fixed-parameter binary logit (Stage 1): adjacent periods merged when a likelihood ratio test does not reject equal parameters at the 90% level. The stage at which the structure was chosen (fixed-parameter or random parameters model) is the one with the lower BIC among models not rejected at the 90% level against the fully unconstrained models (Table 3; all candidates in Table D1). Random parameters (at most eight), heterogeneity in means (at most three terms) and in variances (at most one term) were searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level (a parameter is kept random only while its standard deviation is significant). A variable used in a heterogeneity function enters only the random parameter shown, for crashes in the shared periods, and is removed from the main utility. Marginal effects are average changes in the simulated unconditional probability (2,000 Halton draws) when the indicator switches from 0 to 1, averaged over crashes in the parameter's periods; for random parameters they include the heterogeneity terms. PDO/minor = −(serious/fatal). Percentage above zero evaluated at the heterogeneity variables equal to zero.